

CANADIAN CHEMISTRY OLYMPIAD 2022

(120 minutes)

This segment has five (5) questions. While students are expected to attempt **all** questions for a complete examination in 2 hours, it is recognized that backgrounds will vary and **students will not be eliminated from further competition because they have missed parts of the paper.**

Your answers are to be written in the spaces provided on this paper. All of the paper, is to be returned **immediately** by upload.

— PLEASE READ —

1. BE SURE TO COMPLETE THE INFORMATION REQUESTED AT THE BOTTOM OF THIS PAGE BEFORE BEGINNING THE EXAMINATION.
2. IN QUESTIONS WHICH REQUIRE NUMERICAL CALCULATIONS, BE SURE TO SHOW YOUR REASONING AND YOUR WORK.
3. ONLY NON-PROGRAMMABLE CALCULATORS MAY BE USED ON THIS EXAMINATION.
4. DATASHEET PROVIDED AT THE END OF THIS EXAMINATION IS THE ONLY DATASHEET THAT MAY BE USED ON THIS EXAMINATION.

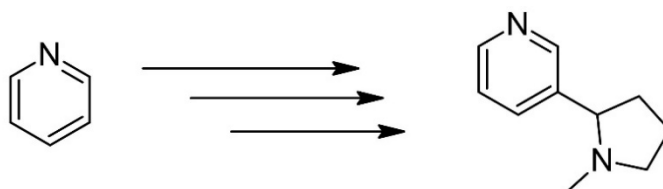
Name _____
(LAST NAME, Given Name; Print Clearly)

School _____

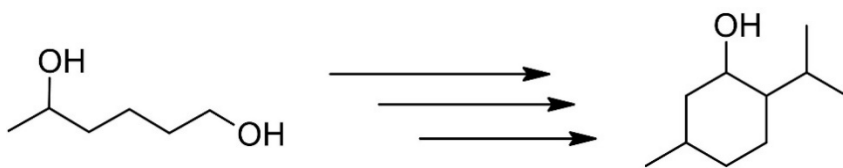
Exam Code _____

1. ORGANIC CHEMISTRY

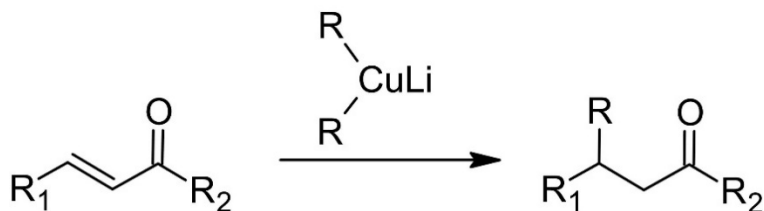
- a) Starting with pyridine and any non-cyclic organic reagents with 6 or less carbon atoms, devise a synthesis of nicotine without stereochemistry. You may use any inorganic reagents you wish. Clearly draw the entire scheme containing reagents and intermediates. **6 marks**



- b) Starting with hexan-1,5-diol and any organic and inorganic reagents you wish, devise a synthesis of menthol without stereochemistry. Clearly draw the entire scheme containing reagents and intermediates. **4 marks**

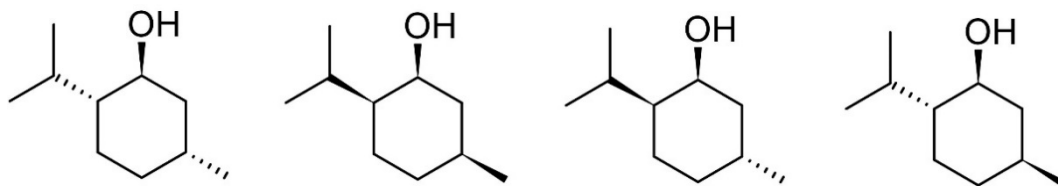


Hint: here's a reaction that may be useful; a **gilman** reagent is a lithium dialkyl cuprate salt that can perform conjugate addition reactions like so:



Where R, R₁ and R₂ are different alkyl groups.

- c) The following structures are all stereoisomers of menthol. Assuming that all these structures are in their most stable conformations, circle the most stable stereoisomer. **2 marks**



2. ANALYTICAL CHEMISTRY

Colorless crystal **A** undergoes a thermal decomposition reaction to produce two gases **B** and **C**. When gas **B** is further heated to a higher temperature and then cooled down to the original temperature, the volume the gases increase by 50%. Although **A** is commonly used in agriculture as a fertilizer, it nevertheless is an oxidizing agent. **A** dissolve easily in water and causes the temperature of the solution to decrease noticeably and the resulting solution is slightly acidic (pH between 4.5 and 5.0). Heating equal moles of **A** and solid NaOH produces a gas **D** with unpleasant odor and a white solid **E**. When gas **D** is introduced into a AgNO_3 solution, a dark brown solid **F** is formed. However, when gas **D** is continuously introduced, a colorless solution is obtained. Heating solid **E** produces colorless gas **G** which is essential for combustion reactions and a white solid **H**. When **H** is treated with concentrated nitric acid, a brown color gas is evolved.

- a) Based on information given, please identify **A**, **B**, **C**, **D**, **E**, **F**, **G** and **H**.
4 marks

A:

B:

C:

D:

E:

F:

G:

H:

- b) Write the chemical reaction equations for the following
4 marks

Reaction to produce **B** & **C**

Reaction for heating **B** to increase the volume by 50%

Reaction to produce **D** & **E**

Reaction to produce **F**

Reaction of **F** to produce the colorless solution

Reaction of **E** to produce **G** and **H**

Reaction of **H** to produce the brown color gas

Leucine $(\text{CH}_3)_2\text{CHCH}_2\text{CH}(\text{NH}_2)\text{COOH}$ is on the top list of essential amino acids for human body. Leucine contains a carboxylic acid functional group and an amine functional group and has a $\text{pK}_a = 2.36$ and $\text{pK}_b = 4.40$. Leucine has been used in the food industry and as healthy supplement.

- c) Using your knowledge of Charge Balance and/or Mass Balance, calculate the pH of a 0.100M aqueous Leucine solution. Show your detailed work to earn full marks.

2.5 marks

In a lab, there are 0.100M NaOH, 0.120M HCl, oxalic acid primary standard ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$, 126.07g/mol), Potassium Hydrogenphthalate primary standard ($\text{KHC}_8\text{H}_4\text{O}_4$, 204.22 g/mol), Tris(hydroxymethyl)-aminomethane primary standard (**Tris**, $(\text{HOCH}_2)_3\text{CNH}_2$, $k_b = 1.15 \times 10^{-6}$, 121.14g/mol), Sodium Carbonate primary standard (Na_2CO_3 , 155.99g/mol), and three indicators, Phenolphthalein ($\text{pK}_a = 9.4$), methyl orange ($\text{pK}_a = 3.4$) and methyl red ($\text{pK}_a = 4.95$).

- d) The purity of Leucine, which is going to be used in making dietary supplement, is to be determined by titration. A 2.000g of Leucine is taken to make a 250.00mL aqueous solution. Which of the afore listed chemicals would you use as the titrant? Which would you use as the indicator?

0.5 mark

- e) Which of the primary standards would you use to standardize your titrant?

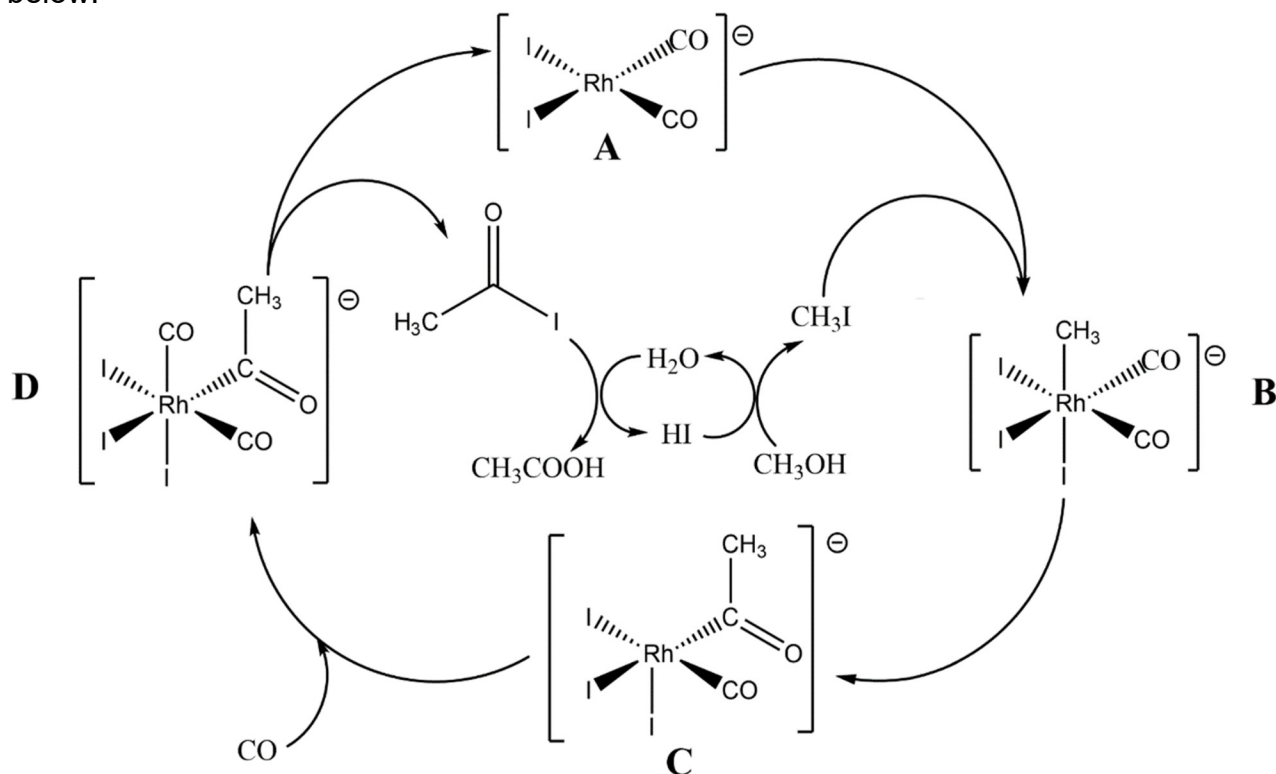
0.5 mark

- f) If 14.94mL of the titrant is required to reach the equivalence point for a 25.00mL aliquot of the analyte, what is the purity of the Leucine sample?

0.5 mark

3. INORGANIC CHEMISTRY

The Monsanto process is a famous industrial catalytic cycle. The process is presented below:



Please answer the following questions pertaining to the Monsanto process:

- Write the *overall* balanced equation for the Monsanto process.
1 mark
- For complex A, state which of its ligands are weak field and which are strong field, and also state whether the complex is a cis or trans isomer.
1 mark
- For complex B, draw its crystal field splitting diagram, making sure to fill in the electrons and label each d orbital. *Hint: complex B is diamagnetic.*
2 marks

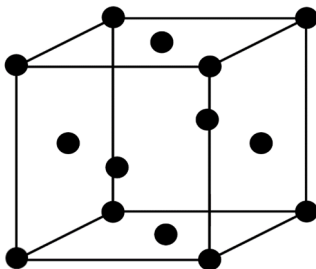
d) For complex C, state its geometry and coordination number.

1 mark

e) For complex D, state the metal's oxidation state and d-electron count.

1 mark

Rhodium, the metal used in the Monsanto process, crystallizes into the face centered cubic structure as shown below:



The lattice parameter (unit cell length) of the crystal is 0.380 nm.

f) State the number of atoms present in the unit cell.

0.5 mark

g) State the coordination number of Rh in the crystal.

0.5 mark

h) Calculate the density of Rh in g cm^{-3} .

2 marks

i) Calculate the volume of empty space in the unit cell of Rh in nm^3 .

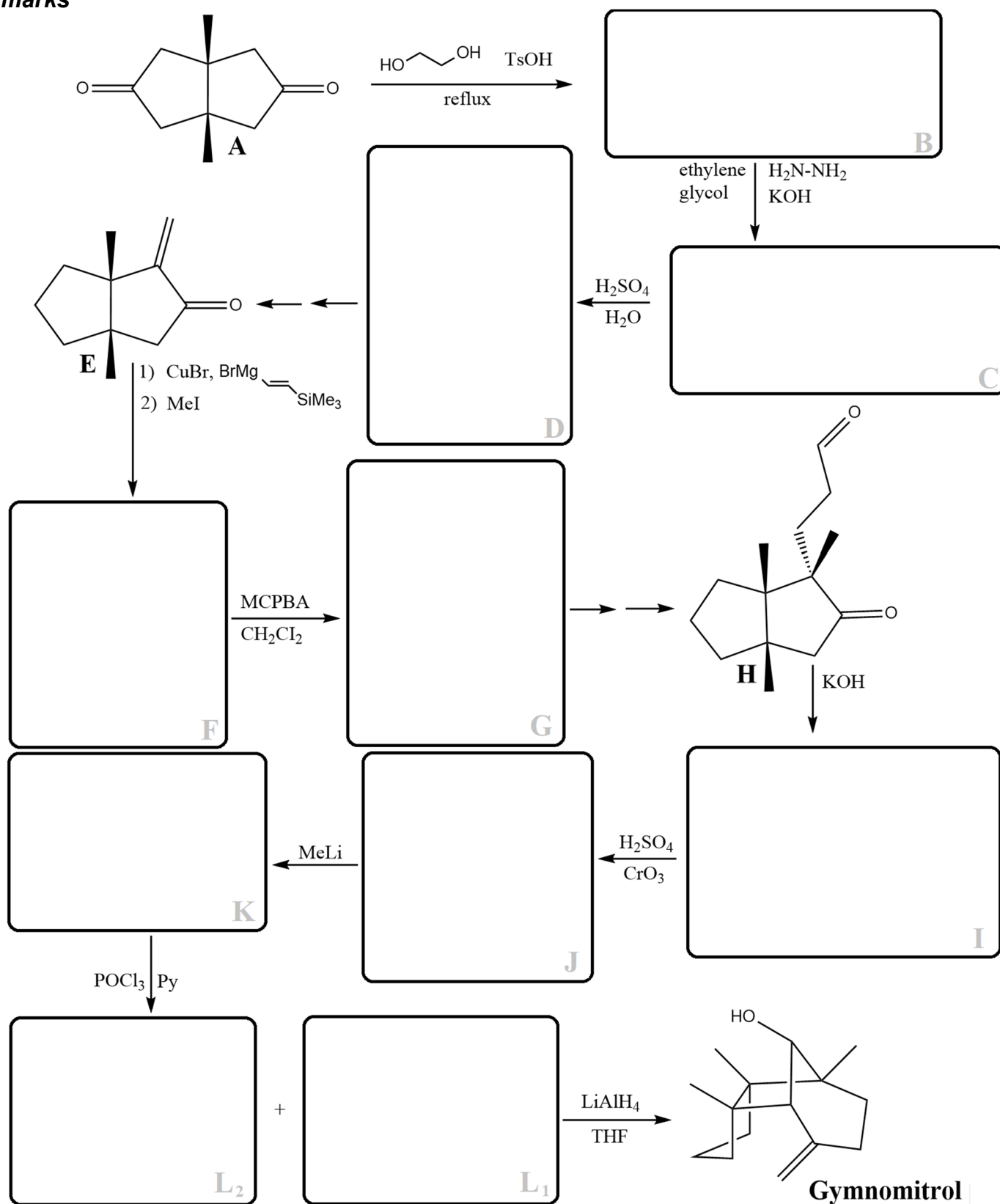
Hint: the volume of a sphere with radius r is given by: $V = \frac{4}{3}\pi r^3$

3 marks

4. ORGANIC CHEMISTRY and NMR spectroscopy

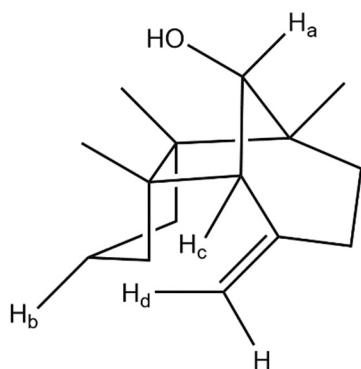
- a) The total synthesis of **Gymnomitrol** combines a wide variety of synthesis techniques. In step 1, only one side is reacted. Over the reaction sequence, a Michael addition and enolate attack are performed consecutively. Later in the sequence, an aldol addition is used to further cyclize the molecule. Given starting compound **A** and the following reaction sequence, identify compounds **B**, **C**, **D**, **F**, **G**, **I**, **J** and **K**. Structures **L**₁ and **L**₂ are both possible products from precursor **K**. Draw them both and note which one reacts to form Gymnomitrol.

9 marks



- b) Step 2 in the synthesis of **Gymnomitrol** is known as a Wolff-Kishner Reduction. Draw the complete reaction mechanism.
1 mark

- c) The selected hydrogen atoms all appear in the condensed H-NMR spectrum. Fill in the table with the hydrogen atoms' corresponding H-NMR peaks.
2 marks



Gymnomitrol

Chemical shift options for peaks:
 5.00, 3.72, 1.65 and 2.53 ppm

Hydrogen atom	Chemical shift of peak (ppm)
a.	
b.	
c.	
d.	

5. PHYSICAL CHEMISTRY

The following 2nd-order reaction: $A(g) \rightarrow 2B(g)$ was carried out at $T = 27\text{ }^{\circ}\text{C}$ in a reaction vessel of constant volume. At the beginning of the reaction, only A(g) at $P = 1\text{ atm}$ was present. After 100 minutes of reaction, the total pressure P in the vessel reaches 1.5 atm. Assume that both A(g) and B(g) are ideal gases.

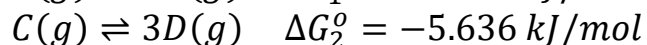
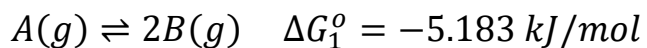
- a) Determine the half-life $t_{1/2}$ and the rate constant k of the reaction at $27\text{ }^{\circ}\text{C}$. State your units in atmosphere and minutes.

3 marks

- b) Give the rate constant k using moles, litres and seconds for the units.

1 mark

Consider a closed container of fixed size in contact with its surroundings maintained at a temperature of 298 K. The inside of this container is partitioned by a frictionless, movable wall into two compartments labeled 1 and 2, with initial volumes of $V_1 = 5L$ and $V_2 = 1L$, respectively. In compartment 1, there is a gaseous equilibrium mixture of molecules A and B with a total pressure of 1 atm. In compartment 2, there is a gas of only compound C also with a pressure of 1 atm. A piece of metal catalyst of negligible volume is then introduced into compartment 2 which causes gas C to decompose into gaseous product D in an equilibrium reaction. This pushes the wall against compartment 1, which increases V_2 and decreases V_1 , also shifting the $A \rightleftharpoons B$ equilibrium as according to Le Chatelier's principle. The wall is pushed until the reactions in both compartments reach a new state of equilibrium. The standard changes in Gibbs free energies for the two equilibria are:



Assume all gases are ideal.

- c) Calculate the initial number of moles for C.
1 mark

- d) Calculate the equilibrium constant for reaction 1 and 2.
1 mark

- e) Calculate the initial number of moles for A and B.
2 marks

We know that V_2 will increase and V_1 will decrease. To get a better idea of how the system may evolve we can define V_{max} , the maximum volume of compartment 2 and V_{min} the minimum volume of compartment 1. To answer f) and g) assume that both compartments are independent from one another and the sum of their volumes is not restricted.

- f) Calculate the value of V_{max} , the maximum volume of compartment 2 at 1 atm.
1 mark

g) Calculate the value of V_{min} , the minimum volume of compartment 1 at 1 atm.

1 mark

h) Once a new state of equilibrium is reached the pressure of the system has changed and the volume of compartment 1 reach 4L. Determine the value of the new equilibrium pressure in the container.

2 marks

-END OF PAPER-



Data Sheet

Fiche de données

1 H 1.008												13	14	15	16	17	18 He 4.003
3 Li 6.941	4 Be 9.012	Relative Atomic Masses (2012, IUPAC) *For the radioactive elements the atomic mass of an important isotope is given										5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3	4	5	6	7	8	9	10	11	12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.96	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.9	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (269)	111 Rg (272)	112 Cn (285)	113 Nh (284)	114 Fl (289)	115 Mc (288)	116 Lv (292)	117 Ts (294)	118 Og (294)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa (231.0)	92 U (238.0)	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Symbol Symbole

Value Quantité numérique

Atomic mass unit	<i>amu</i>	1.66054 x 10 ⁻²⁷ kg	Unité de masse atomique
Avogadro's number	<i>N_A</i>	6.022 x 10 ²³	Nombre d'Avogadro
Charge of an electron	<i>e</i>	1.60218 x 10 ⁻¹⁹ C	Charge d'un électron
Dissociation constant (H ₂ O)	<i>K_w</i>	1.00 x 10 ⁻¹⁴ (25°C)	Constante de dissociation de l'eau (H ₂ O)
Faraday's constant	<i>F</i>	96 485 C mol ⁻¹	Constante de Faraday
Gas constant	<i>R</i>	8.31451 J K ⁻¹ mol ⁻¹ 0.08206 L atm K ⁻¹ mol ⁻¹	Constante des gaz
Mass of an electron	<i>m_e</i>	9.10939 x 10 ⁻³¹ kg	Masse d'un électron
Mass of a neutron	<i>m_n</i>	1.67493 x 10 ⁻²⁷ kg	Masse d'un neutron
Mass of a proton	<i>m_p</i>	1.67262 x 10 ⁻²⁷ kg	Masse d'un proton
Planck's constant	<i>h</i>	6.62608 x 10 ⁻³⁴ J s	Constante de Planck
Speed of light	<i>c</i>	2.997925 x 10 ⁸ m s ⁻¹	Vitesse de la lumière
Rydberg constant	<i>R_H</i>	1.096 x 10 ⁷ m ⁻¹	Constante de Rydberg

1 Å	= 1 x 10 ⁻¹⁰ m
1 atm	= 101.325 kPa
1 bar	= 1 x 10 ⁵ Pa

STP/TPN	SATP/TPAN
273.15 K	298 K
100 kPa	100 kPa